



Fig. 25: FA-18 combat aircraft on the carrier approaching land at dusk (centreline lights are visible and a tiny green spot on the left side is FLOLS)



Fig. 26: AN/SPN-41 radars highlighted (the top-left highlighted section is the LSO station)

It is used for identifying incoming aircraft and operates as a sort of back-end for marshalling operations, whereas the radar operates as the front-end. Every American combat aircraft has a transponder. This transponder is queried by DAIR through AN/SPN-43 radar.

These transponders give their identity and altitude of the aircraft to DAIR through AN/SPN-43 radar. Then, DAIR superimposes this data as alphanumeric over the blips of corresponding aircraft appearing on display consoles. So mere anonymous blips are replaced by names and their altitudes. This system is also capable of processing and su-

perimposing flight plans, geographic reference points and map lines for enhancing mission planning.

Marshalling radar-DAIR combo is a high-precision system enabling controllers to track every aircraft effectively. Through their consoles, air-traffic controllers can interact with DAIR for tracking a particular aircraft's flight. Then they give instructions for the pilot through separate communication channels that exist between air-traffic controllers and pilots. They line up aircraft in a queue for landing and then transfer the control to the landing controllers.

In CATCC, there are many personnel, each handling a specific control-

ling and monitoring task dealing with outbound aircraft, incoming aircraft, landing aircraft, civilian and neutral flights, unidentified blips and the like.

Landing: the mortal challenge

For any pilot, landing his aircraft smoothly is a challenge. But normal pilots come nowhere near their naval counterparts. Naval pilots have to land their aircraft on a moving aircraft carrier, which is the supreme challenge comprising layers of challenges for a pilot's airmanship because of the landing procedure. So naval pilots are considered a special breed among pilots.

Normally, an aircraft after landing runs over the runway, sheds off the enormous momentum, slows down and stops. But the flight-deck of a carrier is not even one-tenth the length of a runway. So the aircraft has to be arrested and trapped to stop its momentum. All aircraft belonging to carriers have a hook, known as a tail hook, beneath their tails. During landing, pilots touch the runway such that the hook snags a thick arresting cable present on the flight deck. The pilot has an error margin of just centimetres. If missed, the pilot increases power, takes off and makes another attempt.

Now comes the next aspect of the challenge—the runway is not only short but also unstable. It pitches up/down and rolls left/right. The pilot has to land guessing how the runway will move and then land, accordingly. The pilot has to guess that in just fractions of a second. During landing, the pilot first aligns the aircraft to the centreline of the runway from an altitude and gradually descends at a constant speed in a shallow angle; it is called approach. In the meantime, if the flight-deck pitches up, the pilot nudges the throttle to increase speed and hence altitude, and vice-versa. Similarly, if the carrier yaws, the pilot adjusts the aircraft, accordingly, and may have to make more than 100 such adjustments during the approach.